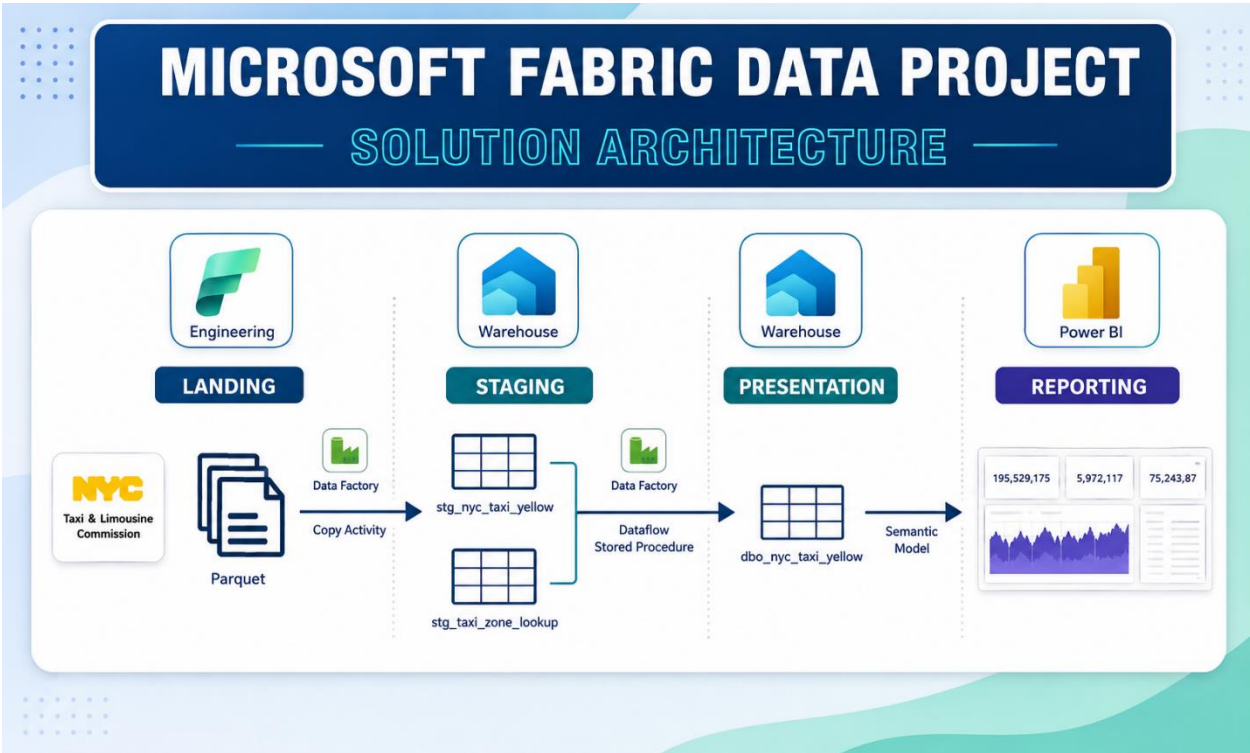


Project Architecture



Created a Fabric Workspace

Fabric | Search | Trials activated: 21 days left

NYCTaxiDataProject | Create deployment pipeline | Create app | Manage access | Workspace settings

+ New item | New folder | Import | Migrate

Filter by keyword | Filter

Choose from predefined task flows or add a task to build one
Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

Select a predefined task flow | Add a task | Import a task flow

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
NYC Taxi Data Pipelines		Folder	—	—	—	—	—	—	
df_pres_processing_nyctaxi		Dataflow Gen...	—	SudFabric	—	—	—	—	
NYC Taxi Project Model		Semantic model	—	NYCTaxiDataPr...	5/22/2026, 10:36...	N/A	—	—	
NYC Yellow Taxi Report		Report	—	NYCTaxiDataPr...	5/22/2026, 10:36...	—	—	—	No
ProjectLakehouse		Lakehouse	—	SudFabric	—	—	—	—	
ProjectLakehouse		SQL analytics ...	—	SudFabric	—	—	—	—	
ProjectWarehouse		Warehouse	—	SudFabric	—	—	—	—	

Data Pipelines created in this Workspace

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

NYTaxiDataProject

...

NYTaxiDataProject

+ New item

New folder

→ Import

↕ Migrate

Create deployment pipeline

Create app

Manage access

Workspace

Filter by keyword

Filter

Choose from predefined task flows or add a task to build one

Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

Select a predefined task flow

Add a task

→ Import a task flow

NYTaxiDataProject > NY Taxi Data Pipelines

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
pl_orchestrate_nyctaxi		Pipeline	—	SudFabric	—	—	—	—	
pl_pres_processing_nyctaxi		Pipeline	—	SudFabric	—	—	—	—	
pl_stg_lookup		Pipeline	—	SudFabric	—	—	—	—	
pl_stg_processing_nyctaxi		Pipeline	—	SudFabric	—	—	—	—	

Landing the data in lakehouse

Created a Lakehouse and uploaded the data files in it.

Fabric

ProjectLakehouse

Search

Trials activated: 21 days left

Analyze data with

Share

Home

Materialized lake views

Get data

New semantic model

Add to data agent

Manage OneLake security

Update all variables

Explorer

Add lakehouses

ProjectLakehouse

Tables

Files

nyctaxi_lookup_zones

nyctaxi_yellow

nyctaxi_yellow

NYTaxiDataProject > ProjectLakehouse > Files > nyctaxi_yellow

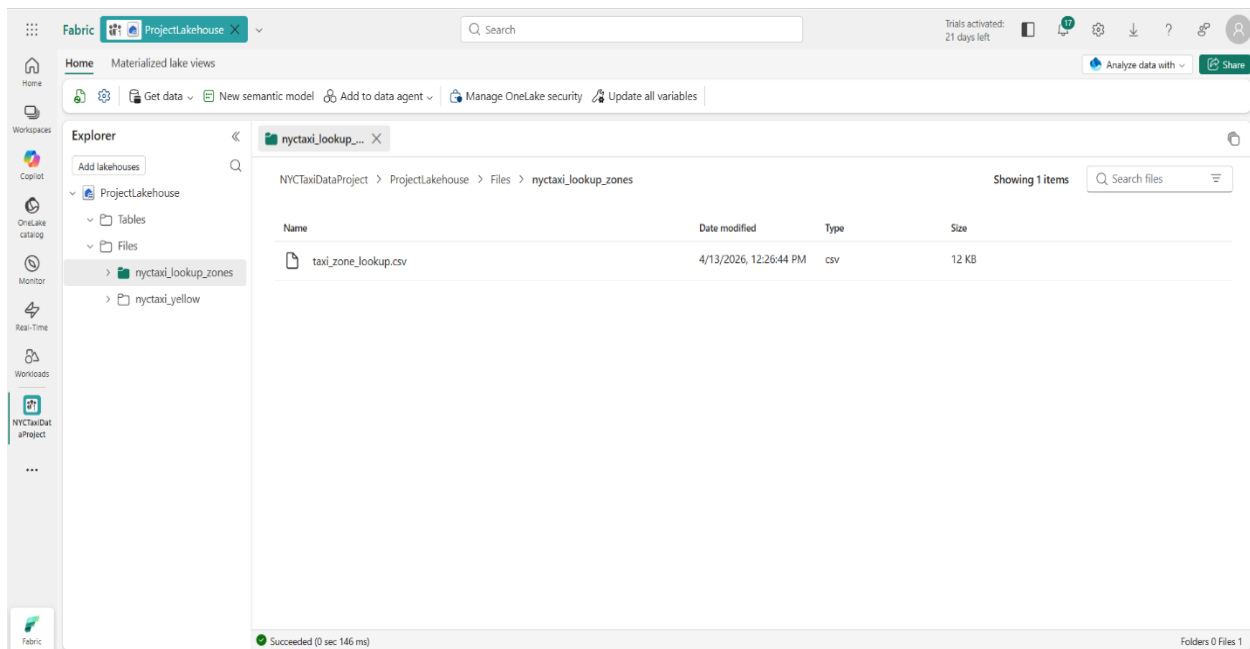
Showing 5 items

Search files

Name	Date modified	Type	Size
yellow_tripdata_2024-01.parquet	4/13/2026, 12:10:19 PM	parquet	47 MB
yellow_tripdata_2024-02.parquet	4/13/2026, 12:10:19 PM	parquet	48 MB
yellow_tripdata_2024-03.parquet	4/13/2026, 12:10:38 PM	parquet	57 MB
yellow_tripdata_2024-04.parquet	4/13/2026, 12:10:38 PM	parquet	56 MB
yellow_tripdata_2024-05.parquet	4/13/2026, 12:10:50 PM	parquet	59 MB

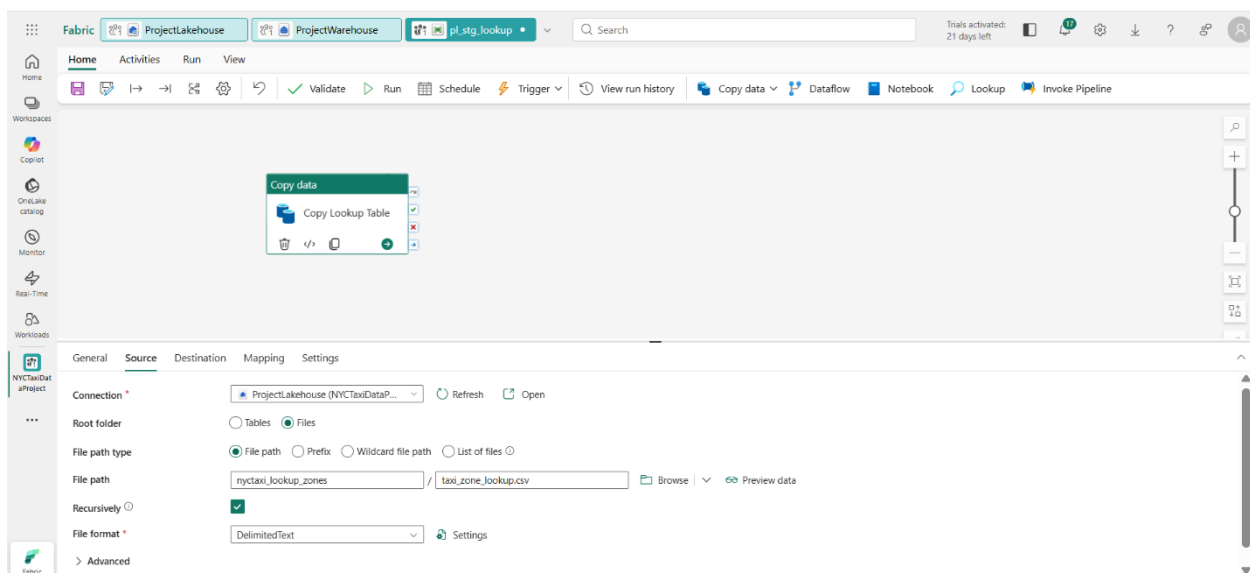
Succeeded (0 sec 133 ms)

Folders 0 Files 5



Steps – From Landing to Staging Presentation layer

Created a data pipeline “**pl_stg_lookup**” with copy activity to copy lookup data from lakehouse to staging table in stg schema in the data warehouse.



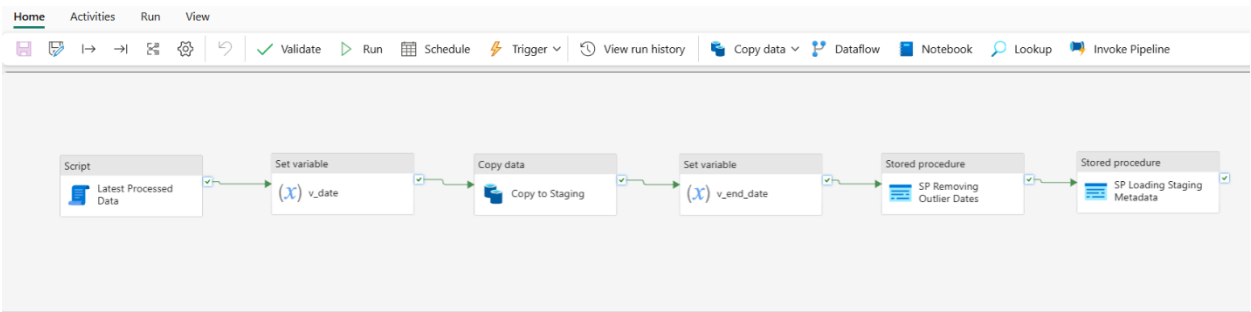
Below is the copied lookup data as it appears in the data warehouse as a staging table under the stg schema after running the pipeline above.

The screenshot shows the Microsoft Fabric ProjectLakehouse interface. On the left, the Explorer pane displays the ProjectWarehouse structure, including Schemas (dbo, INFORMATION_SCHEMA, metadata, queryinsights, stg) and Tables (nyctaxi_yellow, taxi_zone_lookup). The main area shows a 'Data preview - taxi_zone_lookup' table with 1000 rows displayed. The table has four columns: LocationID, Borough, Zone, and service_zone. The data includes various locations like EWR, JFK Airport, LaGuardia Airport, and various Manhattan zones.

LocationID	Borough	Zone	service_zone
1	EBR	Newark Airport	EWR
2	Queens	JFK Airport	Airports
3	Queens	LaGuardia Airport	Airports
4	Unknown	N/A	N/A
5	N/A	Outside of NYC	N/A
6	Manhattan	Alphabet City	Yellow Zone
7	Manhattan	Battery Park	Yellow Zone
8	Manhattan	Battery Park City	Yellow Zone
9	Manhattan	Bloomingdale	Yellow Zone
10	Manhattan	Central Park	Yellow Zone
11	Manhattan	Chinatown	Yellow Zone
12	Manhattan	Clinton East	Yellow Zone
13	Manhattan	Clinton West	Yellow Zone
14	Manhattan	East Chelsea	Yellow Zone
15	Manhattan	East Village	Yellow Zone
16	Manhattan	Financial District North	Yellow Zone
17	Manhattan	Financial District South	Yellow Zone
18	Manhattan	Flatiron	Yellow Zone
19	Manhattan	Garment District	Yellow Zone
20	Manhattan	Governor's Island/Flic Island/Sheerly Island	Yellow Zone

Created pipeline “**pl_stg_processing_nyctaxi**” to ingest taxi trips data from lakehouse to datawarehouse under stg schema.

Code used in **Pipeline: pl_stg_processing_nyctaxi** is as below



Latest Processed Data

For the Script Activity “Latest Processed Data”

```
select top 1 latest_processed_pickup
from metadata.processing_log
where table_processed = 'staging_nyctaxi_yellow'
order by latest_processed_pickup desc;
```

v_date

Pipeline expression for v_date Set Variable activity

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@formatDateTime(addToTime(activity('Latest Processed Data').
output.resultSets[0].rows[0].latest_processed_pickup, 1,
'Month'), 'yyyy-MM')
```

[Clear contents](#)

[Evaluate expression](#)

Activity outputs

Parameters

System variables

Trigger parameters

...

Latest Processed Data

Latest Processed Data activity output

Copy to Staging

Pre Copy Script

Copy command settings

Default values

[+ New](#)

[✖ Failed](#) [More](#)

Pre-copy script ⓘ

```
delete from stg.nyctaxi_yellow
```

v_end_date

Pipeline expression for v_end_date Set Variable activity

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@addToTime(concat(variables('v_date'), '-01'), 1, 'Month')
```

[Clear contents](#)

[Evaluate expression](#)

Activity outputs

[Parameters](#)

[System variables](#)

[Trigger parameters](#)

[...](#)

Copy to Staging

Copy to Staging activity output

Latest Processed Data

Latest Processed Data activity output

SP Removing Outlier Dates

For the Stored Procedure Activity “SP Removing Outlier Dates”.

Create the Stored Procedure **stg.data_cleaning_stg** in the Data Warehouse using the code below.

ProjectWarehouse

Schemas

dbo

INFORMATION_SCHEMA

metadata

queryinsights

stg

Tables

Views

Functions

Stored Procedures

data_cleaning_stg

sys

Preview Copilot uses AI. Mistakes can happen. Verify code suggestions before running them. [Review terms](#)

```
1
2
3 CREATE procedure stg.data_cleaning_stg
4 @end_date DATETIME2,
5 @start_date DATETIME2
6 as
7 delete from stg.nyctaxi_yellow where tpep_pickup_datetime < @start_date or tpep_pickup_datetime > @end_date
```

General
Settings

Connection *
ProjectWarehouse (NYCTaxiDataP...
Refresh
Open

Stored procedure name *
[stg].[data_cleaning_stg]
Refresh

Stored procedure parameters ⓘ

Import
New
Delete

☐	Name	Type *	Value
☐	end_date	DateTime	@variables('v_end_date')
☐	start_date	DateTime	@concat(variables('v_date'), '-01')

SP Loading Staging Metadata

For the Stored Procedure Activity “SP Loading Staging Metadata”.

Code to create the metadata.processing_log table.

ProjectWarehouse
Schemas
dbo
INFORMATION_SCHEMA
metadata
Tables
processing_log
Views

Preview Copilot uses AI. Mistakes can happen. Verify code suggestions before running them. [Review terms](#)

```

1
2
3 create schema metadata;
4
5 create table metadata.processing_log
6 (
7     pipeline_run_id varchar(255),
8     table_processed varchar(255),
9     rows_processed INT,
10    latest_processed_pickup datetime2(6),
11    processed_datetime datetime2(6)
12 );
13

```

Created the Stored Procedure metadata.insert_staging_metadata in the Data Warehouse using the code below.

ProjectWarehouse

Schemas

dbo

INFORMATION_SCHEMA

metadata

Tables

Views

Functions

Stored Procedures

insert_presentation_metadata

insert_staging_metadata

Preview

Copilot uses AI. Mistakes can happen. Verify code suggestions before running them. [Review terms](#)

```

1
2 CREATE PROCEDURE metadata.insert_staging_metadata
3     @pipeline_run_id VARCHAR(255),
4     @table_name VARCHAR(255),
5     @processed_date DATETIME2
6 AS
7     INSERT INTO metadata.processing_log (pipeline_run_id, table_processed, rows_processed, latest_processed_pickup, processed_datetime)
8     SELECT
9         @pipeline_run_id AS pipeline_id,
10        @table_name AS table_processed,
11        COUNT(*) AS rows_processed,
12        MAX(tpep_pickup_datetime) AS latest_processed_pickup,
13        @processed_date AS processed_datetime
14    FROM stg.nyctaxi_yellow;

```

General

Settings

Connection *

ProjectWarehouse (NYCTaxiDataP...

Refresh

Open

Stored procedure name *

[metadata].[insert_staging_metadata]

Refresh

Stored procedure parameters ⓘ

← Import

+ New

🗑 Delete

☐	Name	Type *	Value
☐	pipeline_run_id	String	@pipeline().RunId
☐	processed_date	DateTime	@pipeline().TriggerTime
☐	table_name	String	staging_nyctaxi_yellow ☐ Treat as null

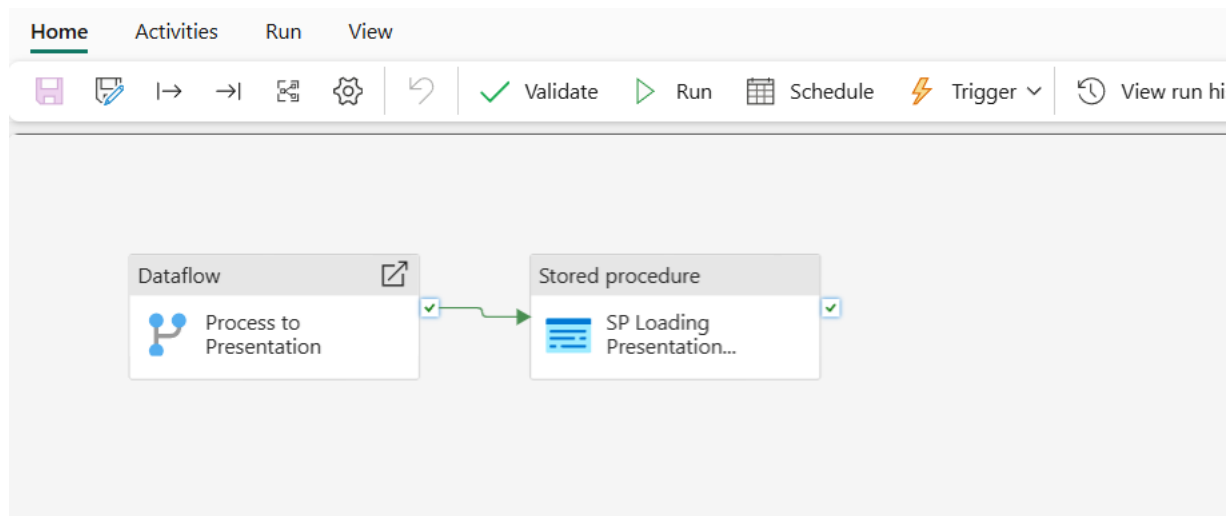
Add dynamic content [Alt+Shift+D]

Steps - From Staging to Presentation layer

Overall Pipeline

Code used in Pipeline: pl_pres_processing_nyctaxi

This is a view of the pipeline using the Dataflow Gen2 and a Stored Procedure activity.



Create the `dbo.nyctaxi_yellow` table

This is the initial empty table so we can load the data from the Dataflow into this table.

ProjectWarehouse

Schemas

dbo

Tables

nyctaxi_yellow

Views

Functions

Stored Procedures

INFORMATION_SCHEMA

metadata

queryinsights

stg

Preview

Copilot uses AI. Mistakes can happen. Verify code suggestions before running them. [Review terms](#)

```
1
2
3 CREATE TABLE dbo.nyctaxi_yellow
4 (
5     vendor varchar(50),
6     tpep_pickup_datetime date,
7     tpep_dropoff_datetime date,
8     pu_borough varchar(100),
9     pu_zone varchar(100),
10    do_borough varchar(100),
11    do_zone varchar(100),
12    payment_method varchar(50),
13    passenger_count int,
14    trip_distance FLOAT,
15    total_amount FLOAT
16 );
17
18
19
20
```

Dataflow - Process to Presentation

Created the below dataflow to clean, transform and merge the data from the staging tables and load the final data into the destination table **`dbo.nyctaxi_yellow`** table in the Datawarehouse.

Power Query

Home Transform Add column View Help

Search (Alt + Q)

Queries [4]

- stg nyctaxi_yellow
- stg taxi_zone_lookup
- Merge
- Merge (2)

Table: ReorderColumns("#Removed columns", {"vendor", "tpep_pickup_datetime", "tpep_dropoff_datetime", "pu_borough", "pu_zone", "do_borough", "do_zone"}

	vendor	tpep_pickup_datetime	tpep_dropoff_datetime	pu_borough	pu_zone	do_borough	do_zone
1	Curb Mobility	3/4/2024	3/4/2024	Queens	JFK Airport	Manhattan	Midtown Ce
2	Curb Mobility	3/1/2024	3/1/2024	Queens	JFK Airport	Manhattan	Midtown Ce
3	Curb Mobility	3/13/2024	3/1/2024	Queens	JFK Airport	Manhattan	Midtown Ce
4	Curb Mobility	3/1/2024	3/1/2024	Queens	JFK Airport	Manhattan	Midtown Ce
5	Curb Mobility	3/18/2024	3/18/2024	Queens	JFK Airport	Manhattan	Midtown Ce
6	Curb Mobility	3/20/2024	3/20/2024	Queens	JFK Airport	Manhattan	Midtown Ce
7	Curb Mobility	3/1/2024	3/1/2024	Queens	JFK Airport	Manhattan	Midtown Ce
8	Curb Mobility	3/12/2024	3/12/2024	Queens	JFK Airport	Manhattan	Midtown Ce
9	Curb Mobility	3/5/2024	3/5/2024	Queens	JFK Airport	Manhattan	Midtown So
10	Curb Mobility	3/13/2024	3/13/2024	Queens	JFK Airport	Manhattan	Midtown So

Query settings

Properties

Name

Merge (2)

Applied steps

- Source
- Expanded stg taxi_zone_lookup
- Renamed columns
- Reordered columns
- Removed columns
- Reordered columns 1

Data destination

- Warehouse

SP Loading Presentation Metadata

For the Stored Procedure Activity “SP Loading Presentation Metadata”.

Create the Stored Procedure **metadata.insert_presentation_metadata** in the Data Warehouse using the code below.

+ Warehouses

ProjectWarehouse

- Schemas
 - dbo
 - INFORMATION_SCHEMA
 - metadata
 - Tables
 - Views
 - Functions
 - Stored Procedures
 - insert_presentation_metadata
 - insert_staging_metadata
 - queryinsights

Run Save as Save as view Create rule Explain query Fix query errors

Preview Copilot uses AI. Mistakes can happen. Verify code suggestions before running them. [Review terms](#)

```

1
2
3 CREATE PROCEDURE metadata.insert_presentation_metadata
4     @pipeline_run_id VARCHAR(255),
5     @table_name VARCHAR(255),
6     @processed_date DATETIME2
7 AS
8     INSERT INTO metadata.processing_log (pipeline_run_id, table_processed, rows_processed, latest_processed_pickup, processed_datetime)
9     SELECT
10         @pipeline_run_id AS pipeline_id,
11         @table_name AS table_processed,
12         COUNT(*) AS rows_processed,
13         MAX(tpep_pickup_datetime) AS latest_processed_pickup,
14         @processed_date AS processed_datetime
15     FROM dbo.nyctaxi_yellow;
16
17
18
19
20

```

General
Settings

Connection *
ProjectWarehouse (NYCTaxiDataP...
Refresh
Open

Stored procedure name *
[metadata].[insert_presentation_met...
Refresh

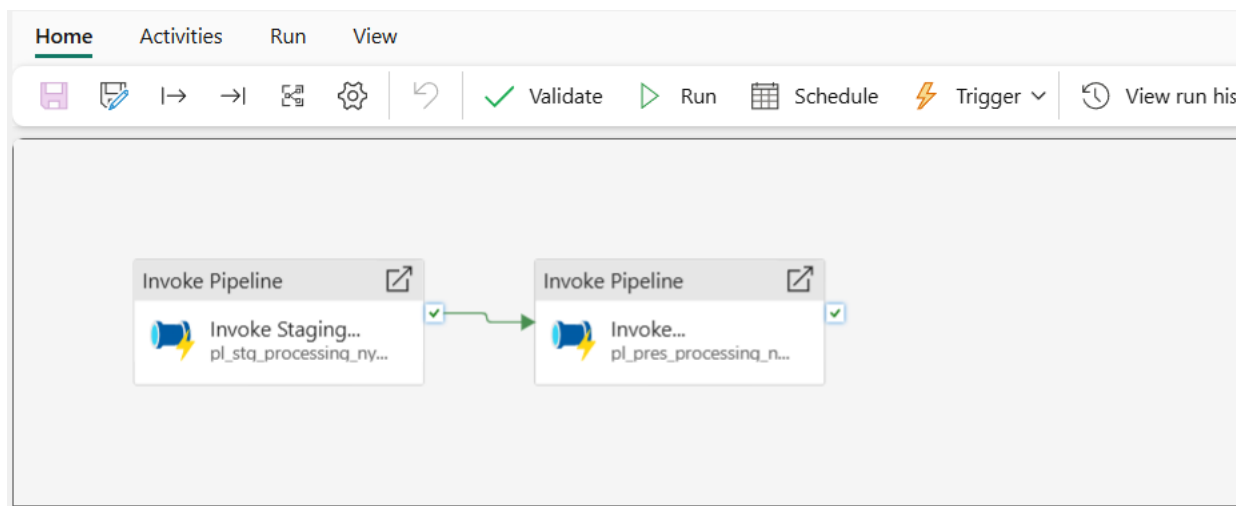
Stored procedure parameters ⓘ

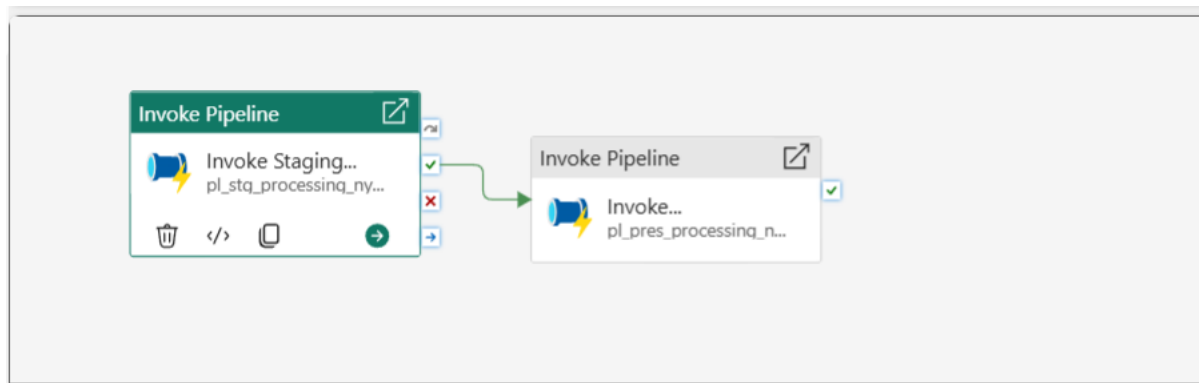
Import
New
Delete

☐	Name	Type *	Value
☐	pipeline_run_id	String	@pipeline().RunId
☐	processed_date	DateTime	@pipeline().TriggerTime
☐	table_name	String	presentation_nyctaxi_y... ☐ Treat as null

Steps – End-To-End Data Processing

Created an orchestration pipeline “**pl_orchestrate_nyctaxi**” (using “invoke pipeline” activity) that invokes both the pipelines “pl_stg_processing_nyctaxi” and “pl_pres_processing_nyctaxi” that are already created. This orchestration pipeline will run end-to-end data process.





General **Settings**

Type ☒ Fabric ☐ Azure Data Factory ☐ Azure Synapse Analytics

Connection * ⓘ

FabricDataPipelines MyFabric

Refresh

Edit

Workspace *

NYCTaxiDataProject

Refresh

Pipeline *

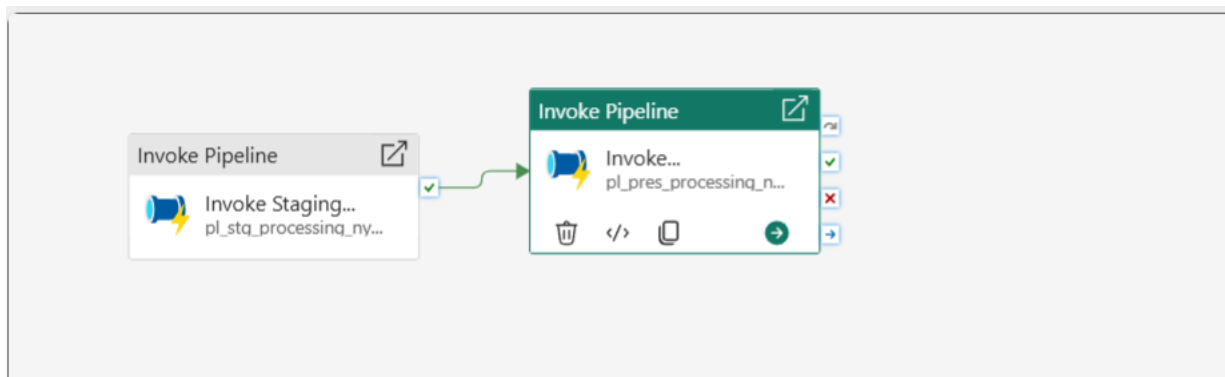
pl_stg_processing_nyctaxi

Refresh

Open

+ New

Parameters



General **Settings**

Type ☒ Fabric ☐ Azure Data Factory ☐ Azure Synapse Analytics

Connection * ⓘ

FabricDataPipelines MyFabric

Refresh

Edit

Workspace *

NYCTaxiDataProject

Refresh

Pipeline *

pl_pres_processing_nyctaxi

Refresh

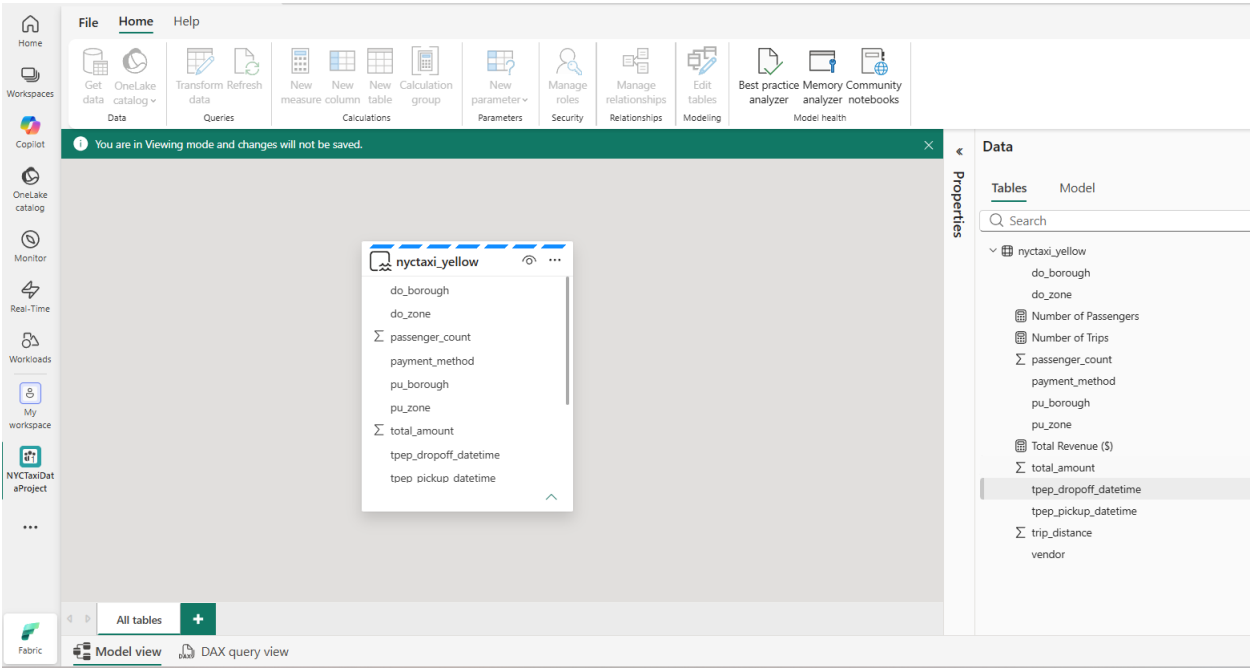
Open

+ New

Parameters

+ New

Semantic Model



Power BI Dashboard

